Unit 6, Lesson 8: Proving Triangle Similarity – Part 1

Benchmark

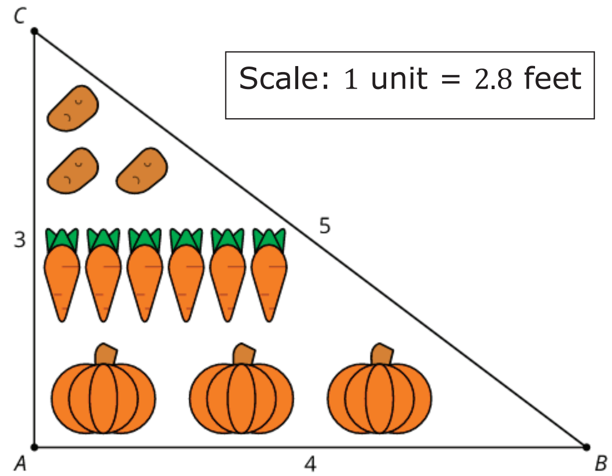
MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

Learning Target

☐ I can use triangle similarity to prove triangles are similar.

6.8.1 Warm-Up: Vegetable Garden

1. ΔABC is used to create a plan for a vegetable garden.



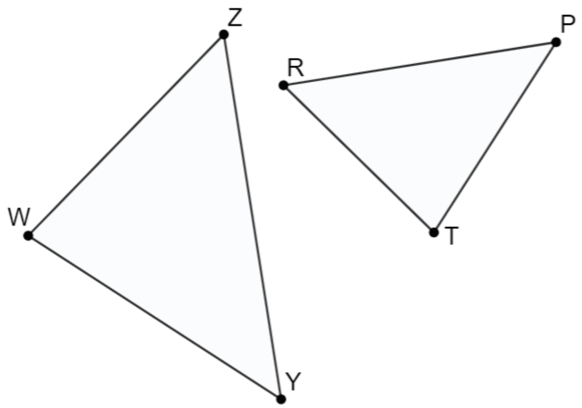
a. What do you notice about the plan?

b. Write at least three equivalent ratios or equations using lengths from both the diagram and the full-size garden.

6.8.2 Collaboration: Proving Triangle Similarity

With your partner, answer the following questions.

1. ΔWZY and ΔRPT are shown. $\angle WZY \cong \angle TRP$ and $\angle ZWY \cong \angle RTP$.



a. Mark the figure using the given information.

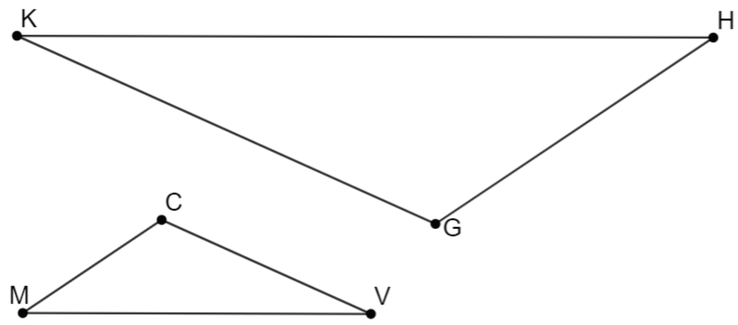
b. What similarity theorem can be applied to show that the triangles are similar?

c. Write the similarity statement.

d. Complete each.

$\frac{WZ}{\_\_\_\_} = \frac{\_\_\_\_}{PT}$ $\qquad$ $\frac{ZY}{\_\_\_\_} = \frac{\_\_\_\_}{RT}$ $\qquad$ $\angle WYZ \cong$

2. ΔKHG and ΔMVC are shown. $\angle KHG \cong \angle VMC$ and $\frac{KH}{VM} = \frac{GH}{CM}$.



a. Mark the figure using the given information.

b. What similarity theorem can be applied to show that the triangles are similar?

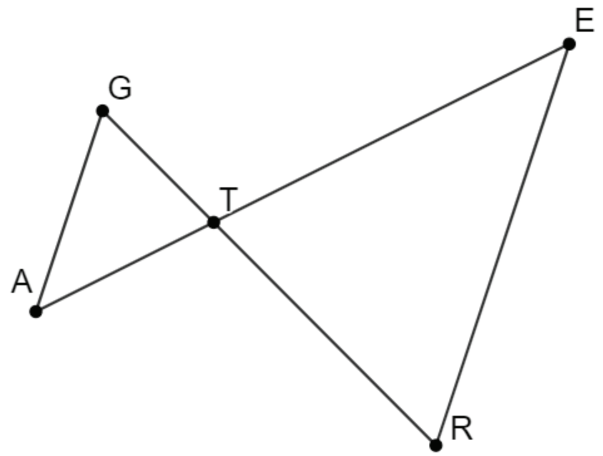
c. Write the similarity statement.

d. Complete each.

$\frac{KG}{\_\_\_\_} = \frac{\_\_\_\_}{CM}$ $\qquad$ $\angle HGK \cong$ $\qquad$ $\angle CVM \cong$

6.8.3 Guided Instruction: Proving Triangle Similarity

1. ΔGTA and ΔTER are shown, and $\overline{GA} \parallel \overline{ER}$.



a. With your partner, discuss what conclusions can be drawn using the given information. Summarize your discussion and explain how you reached each conclusion.

b. Which similarity postulates should be used to prove $\Delta GTA \sim \Delta RTE$?

c. Complete the two-column proof.

Statement	Reason
1. $\overline{GA} \parallel \overline{ER}$	1.
2.	2. If two parallel lines are intercepted by a transversal, then alternate interior angles are congruent.
3.	3.
4. $\Delta GTA \sim \Delta RTE$	4.

d. Ask two classmates about their proof. Record their step 3 statement and reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Step 3 Statement	Step 3 Reason	Initials

e. Discuss with your partner the two different ways that $\Delta GTA \sim \Delta RTE$ can be proved.

f. Explain why you do not need to state three congruent angles in your two-column proof.

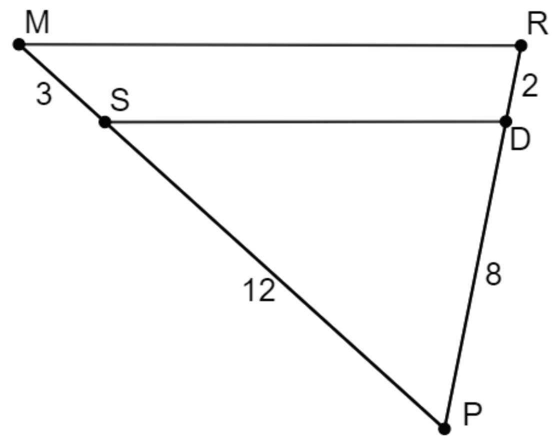
g. Arian and Isabella were discussing why there is no Angle-Angle-Angle Similarity Theorem.

Arian	Isabella
I think once you know two pairs of angles in a triangle are congruent, then you know the last pair are congruent since the sum of all of the angles of a triangle are $180°$.	I think once you know two pairs of angles in a triangle are congruent, then you know all of the sides are proportional.

Discuss with your partner why Arian and Isabella's arguments could also be applied to show why Angle-Side-Angle similarity and Angle-Angle-Side similarity is not needed to prove similarity. Summarize your discussion.

2. ΔSPD and ΔMPR are shown.

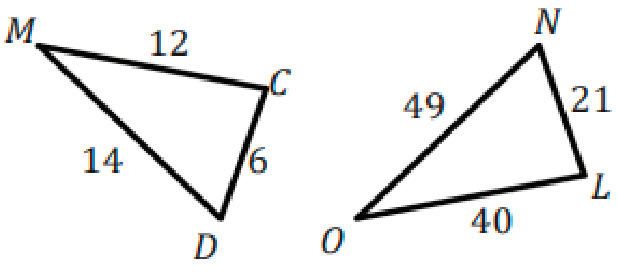
Given: $MS = 3$, $SP = 12$, $RD = 2$, and $DP = 8$
Prove: $\Delta SDP \sim \Delta MRP$



Statement	Reason
1. $MS = 3$, $SP = 12$, $RD = 2$, and $DP = 8$	1.
2. $MS + SP = PM$ $RD + DP = RP$	2.
3. $PM = 15$ $RP = 10$	3. Substitution Property of Equality
4. $\frac{PS}{PM} = \frac{12}{15}$ $\frac{PD}{PR} = \frac{8}{10}$	4.
5. $\frac{PS}{PM} = \frac{PD}{PR}$	5.
6. $\angle P \cong \angle P$	6.
7. $\Delta SDP \sim \Delta MRP$	7.

6.8.4 Try It: Triangle Similarity

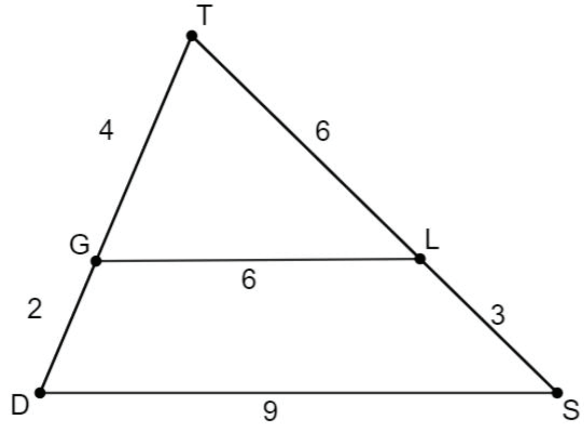
1. Amanda and Tyler are determining if $\Delta MCD \sim \Delta OLN$ is true. Their reasonings are shown.



Amanda's Answer	Tyler's Answer
No, the triangles are not similar because the corresponding sides are not proportional.	Yes, the triangles are similar because the corresponding sides are proportional.

Who do you agree with? Explain your reasoning.

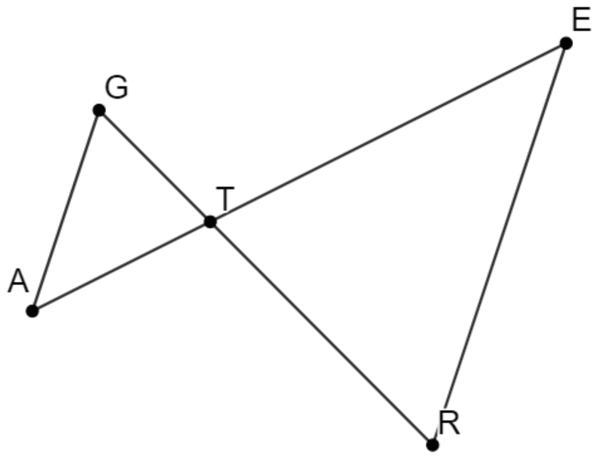
2. ΔTLG and ΔTSD are shown.
Given: $TG = 4$, $GD = 2$, $TL = 6$, $LS = 3$, $GL = 6$, and $DS = 9$
Prove: $\Delta TLG \sim \Delta TSD$



Statement	Reason
1. $TG = 4$, $GD = 2$, $TL = 6$, $LS = 3$, $GL = 6$, and $DS = 9$	1.
2. $TG + GD = TD$ $TL + LS = TS$	2.
3. $TD = 6$, $TS = 9$	3. Substitution
4. $\frac{TG}{TD} = \frac{4}{6}$, $\frac{TL}{TS} = \frac{6}{9}$, $\frac{GL}{DS} = \frac{6}{9}$	4.
5. $\frac{TG}{TD} = \frac{TL}{TS} = \frac{GL}{DS}$	5.
6. $\Delta TLG \sim \Delta TSD$	6.

6.8.5 Wrap-Up: Ticket Out the Door

1. ΔGTA and ΔTER are shown.



Given: $TA = \frac{2}{3}TE$ and $TG = \frac{2}{3}RT$
Prove: $\Delta GTA \sim \Delta RTE$

Complete the two-column proof.

Statement	Reason
1. $TA = \frac{2}{3}TE$ and $TG = \frac{2}{3}RT$	1. Given
2. $\frac{TA}{TE} = \frac{2}{3}$, $\frac{TG}{RT} = \frac{2}{3}$	2.
3.	3. Transitive Property
4. $\angle GTA \cong \angle RTE$	4.
5. $\Delta GTA \sim \Delta RTE$	5.

Unit 6, Lesson 9: Proving Triangle Similarity – Part 2

Benchmark

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Learning Target

☐ I can use triangle similarity to prove that triangles are similar.